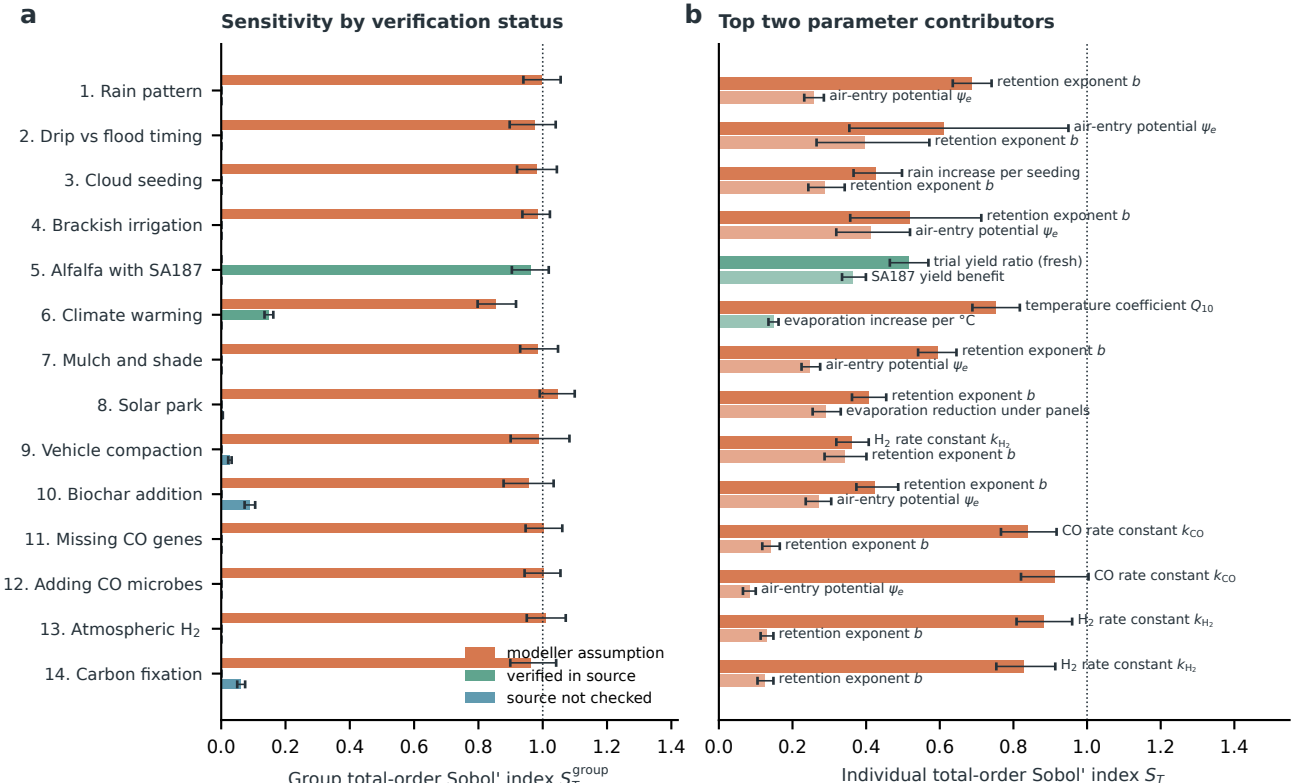
a
Sensitivity by verification status
1. Rain pattern
2. Drip vs flood timing
3. Cloud seeding
4. Brackish irrigation
5. Alfalfa with SA187
6. Climate warming
7. Mulch and shade
8. Solar park
9. Vehicle compaction
10. Biochar addition
11. Missing CO genes
12. Adding CO microbes
13. Atmospheric H_2
14. Carbon fixation
modeller assumption
verified in source
source not checked
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4
Group total-order Sobol' index S_T^{group}
b
Top two parameter contributors
retention exponent b
air-entry potential ψ_e
air-entry potential ψ_e
retention exponent b
rain increase per seeding
retention exponent b
retention exponent b
air-entry potential ψ_e
trial yield ratio (fresh)
SA187 yield benefit
temperature coefficient Q_{10}
evaporation increase per °C
retention exponent b
air-entry potential ψ_e
retention exponent b
evaporation reduction under panels
H_2 rate constant k_{H_2}
retention exponent b
retention exponent b
air-entry potential ψ_e
CO rate constant k_{CO}
retention exponent b
CO rate constant k_{CO}
air-entry potential ψ_e
H_2 rate constant k_{H_2}
retention exponent b
H_2 rate constant k_{H_2}
retention exponent b
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4
Individual total-order Sobol' index S_T